

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I **P. Reddy Suhail** /192312671L (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Suhail

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies	Appropriate use of tools	Limited or inconsistent	Incorrect or minimal use of tools

		s with proper configuration	with correct execution	t use of tools	
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



A COMPARATIVE TECHNICAL STUDY

6G Communication Architecture

Compared with the OSI Model

Explained in simple terms — what it is, how it works, and how it compares

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Introduction

Every network needs rules for how devices talk to each other — just like people need a common language. The OSI Model is a simple, well-known rulebook that splits this job into 7 easy steps. 6G is the next generation of mobile networks: much faster, smarter, and able to “think” using AI. In this presentation, we compare the new 6G design with the classic OSI rulebook to see what stays the same and what changes.

1

What is OSI?

A simple 7-step rulebook that all of today's networks follow to send data.

2

What is 6G?

The next generation of mobile network — super-fast, smart, and AI-powered.

3

Why compare them?

To see which parts of the old rulebook 6G keeps, and which parts it changes.

4

Why it matters

Understanding this helps us see how future technology will work and improve.

02

The OSI Model — Layers 3 to 1

The bottom layers get data physically from one device to another, across the right path.

3

Network

Decides the best path for data to travel across different networks, using logical addresses.

IP · ICMP · Routers

2

Data Link

Moves data directly between two connected devices on the same network, and checks for errors.

Ethernet · Wi-Fi · MAC Address · Switches

1

Physical

The actual hardware — sends raw bits as electrical signals, light pulses, or radio waves.

Cables · Fiber Optics · Radio Waves · Hubs

Data flows top → bottom (Layer 7 → 1) when sending, and bottom → top (Layer 1 → 7) when receiving.

04

